

Kasra Ferdowsi

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Engineering Experience

Staff Software Engineer *Cubist Inc*

2024 - Current

- Contributed over 250 pull requests spanning Rust back-end services, AWS CDK infrastructure, TypeScript and Rust SDKs, WebAssembly policies, example programs, and documentation.
- Designed and built an Access Control (ACL) system in Rust providing fine-grained user-friendly control over sensitive resources such as CubeSigner keys, policies, and secrets.
- Productionized and maintained the Cubist Policy SDK; Building its CI/CD pipeline, fixing bugs and outdated dependencies, and extending it with JSON and gRPC clients and other features.
- Developed the CubeSigner RPC API for transaction management and submission, including fully-concurrent EVM nonce management.
- Led design, requirements gathering, and implementation for the second iteration of the Bascule Drawbridge; Preventing loss of funds due to malicious actors or bugs from the Lombard protocol.

Graduate Student Researcher *UC San Diego*

2019 - 2024

- Invented novel techniques and interaction models for using Bottom-up Enumerative Synthesis for efficient synthesis of Python programs.
- Maintained the Projection Boxes live programming environment, extended it with program synthesis features, and ported it to a web editor with a Java backend.
- Ran multiple user studies with up to hundreds of participants to evaluate the usability of programming tools, and visualized and performed statistical analysis on the resulting study data.

Research Intern *Microsoft Research*

2022

- Designed and implemented ColDeco, a novel interaction model for inspecting and diagnosing errors in AI-generated code by end-users.

Software Engineer Intern *Tom Sawyer Software*

2016 - 2019

- Helped design and develop the Tom Sawyer Licensing Server 2.0, a backwards-compatible scalable web server, in Java using a micro-service architecture.
- Implemented account management, and a web interface for Graph Database Browser.
- Co-designed and developed a tool combining Git, Maven and Docker to streamline building and deploying Tom Sawyer Perspectives applications.

Education

UC San Diego, California, USA

2024

- PhD in Computer Science advised by Nadia Polikarpova and Sorin Lerner.
- M.S. in Computer Science.

UC Berkeley, California, USA

2019

- B.S. in Electrical Engineering and Computer Science.
- Graduated with Highest Honors.

Expertise

Programming Languages

- Advanced knowledge of Rust and Python. Fluent in TypeScript, Scala, and Java. Familiar with Haskell and JavaScript.

DevOps & Infrastructure

- Advanced knowledge of DynamoDB. Familiar with many AWS services (EC2, Lambdas, Secrets Manager, etc.), AWS CDK, GitHub Actions, and Git.

Publications

Bottom-up Synthesis of Memory Mutations with Separation Logic Ferdowsi, K., Peleg, H.	ECOOP 2025
Human-Centered Program Synthesis Ferdowsifard, K.	Dissertation
Validating AI-Generated Code with Live Programming Ferdowsi, K., Huang, R., James, M. B., Polikarpova, N., and Lerner, S.	CHI 2024
ColDeco: An End User Spreadsheet Inspection Tool for AI-Generated Code Ferdowsi, K., Williams, J., Drosos, I., Gordon, A.D., Negreanu, C., Polikarpova, N., Sarkar, A., and Zorn, B.	VL/HCC 2023
Towards Human-Centered Types and Type Debugging Ferdowsi, K.	PLATEAU 2023
The Usability of Advanced Type Systems: Rust as a Case Study Ferdowsi, K.	Research Exam
Investigating the Impact of Using a Live Programming Environment in a CS1 Course Huang, R., Ferdowsi, K., Selvaraj, A., Soosai Raj, A.G., and Lerner, S.	SIGCSE TS 2022
LooPy: Interactive Program Synthesis with Control Structures Ferdowsifard, K., Barke, S., Peleg, H., Lerner, S., and Polikarpova, N.	OOPSLA 2021
Small-Step Live Programming by Example Ferdowsifard, K., Ordookhanians, A., Peleg, H., Lerner, S., and Polikarpova, N.	UIST 2020

Teaching Experience

Teaching Assistant <i>CSE130: Programming Languages</i>	Fall 2023
Teaching Assistant <i>CSE230: Graduate Programming Languages</i>	Fall 2022
Teaching Assistant <i>CSE130: Programming Languages</i>	Winter 2022
Teaching Assistant <i>CSE130: Programming Languages</i>	Fall 2021
Teaching Assistant <i>CSE11: Accelerated Intro to Programming</i>	Spring 2020
Teaching Assistant <i>CSE131: Compiler Construction</i>	Fall 2019

Service

Student Organizer	PLATEAU 2024
External Reviewer	CHI 2023
Program Committee Member	HATRA 2023
Student Volunteer	PLDI 2023

Mentorship

GradWIC Mentor	2022 - 2023
CSE599: Teaching Methods in Computer Science Mentor	Winter 2023
ERSP Mentor	2021 - 2022